L-F-340B
March 29, 1973
SUPERSEDING
Fed. Spec. L-F-340A
May 9, 1967,
MIL-F-9621(USAF)
August 13, 1957, and
MIL-F-4491(USAF)
March 17, 1954

FEDERAL SPECIFICATION

FILM, DIAZOTYPE, SENSITIZED; MOIST AND DRY PROCESS; ROLL AND SHEET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sensitized flexible synthetic resin stock, coated with diazo compounds, in sheet or roll form and which is dimensionally stable within the limits specified herein. This film is intended for use in making prints or intermediates of original copy, drawings, and tracings.

1.2 Classification. The sensitized material covered by this specification shall be classified according to the following types, classes, subclasses and styles, as specified (see 6.2).

Types:

I - Diazotype, ammonia vapor process.
II - Diazotype, moist process.

Classes:

1. Sensitized, matte surface, smooth back.
2. Sensitized, matte surface, matte back.
3. Sensitized, smooth surface, matte back.
4. Sensitized, smooth surface, smooth back.

Subclasses:

a. High dimensional stability.
b. Medium dimensional stability.

Styles:

1A - Black line.
1B - Sepia line.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-I-528	-	Ink, Drawing, Waterproof, Black.
UU-P-221	-	Paper, Direct-Positive Sensitized (Diazotype-Moist And Dry Process).
PPP-B-621	-	Boxes, Wood, Nailed And Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60	-	Tape, Packaging, Waterproof.

FSC 6750

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Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 131 - Diazotype Paper Sheet Sizes.
Fed. Std. No. 406/GEN - Plastics: Methods Of Testing.
406/1013 - Tensile Properties Of Thin Plastic Sheets And Films.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case And Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129 - Marking For Shipment And Storage.
MIL-STD-147 - Palletized And Containerized Unit Load.

(Copies of Military Specifications And Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI), Inc. Standards:

No. PH-1.16 - Graphic Arts Sheet Film (Inch and Centimeter Sizes), Dimensions For.
No. PH-1.30 - Films in Rolls for Recording Instruments, Graphic Arts, Phototypesetting, Portrait, X-Ray, And Related Use; Dimensions For.
No. PH-1.31 - Brittleness of Photographic Film, Method For Determining The.
No. PH-1.32 - Dimensional Change Characteristics of Photographic Films And Papers, Methods For.
No. PH-2.17 - Diffuse Reflc Density.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

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National Bureau of Standards (NBS) Circular Letter:

No. LC-547 - Preparation and Colorimetric Properties of Magnesium Oxide Reflectance Standard.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials. The sensitized material shall be furnished in rolls or sheets, as specified (see 6.2). The material shall be free of included air, wrinkles, blisters, cracks, and substantially free from defects such as scratches, dents, pin holes, pores and heat marks. The sensitized base shall be of high quality and entirely suitable for the purpose intended. The working surface shall be as specified herein and the sheets or rolls shall be essentially clean and free from dirt which would impair the image area. The film shall not be damaged by normal handling. Any material which exhibits physical or coating defects prior to expiration date shall be replaced by the manufacturer at no additional cost to the Government.

3.2 Data requirement. When specified (see 6.2), data as indicated shall be furnished in accordance with the contract or purchase order.

3.3 Thickness. Unless otherwise specified, the thickness of the base shall be no less than 0.002 inches and no greater than 0.0075 inches. The procuring activity shall specify the thickness required. Standard thicknesses in inches are 0.002, 0.003, 0.004, 0.005, 0.007, 0.0025, and 0.0075. Thickness variation are to be within ±10 percent of the standard specified thickness, and shall be measured in accordance with 4.4.

3.4 Coating adhesion. All coatings shall adhere to and shall not strip from the film base during processing. Adhesion shall be such that the coatings of processed sheets will not be removed when tested, as specified in 4.5.

3.5 Coating defects. The sensitized film shall be essentially free from coating defects when tested, as specified in 4.6.

3.6 Tensile strength. The film shall have a tensile strength of at least 10,000 pounds per square inch, in the lengthwise direction, when tested, as specified in 4.7.

3.7 Brittleness. The brittleness of unprocessed sheets shall conform to ANSI PH-1.31 except that the temperature shall be 70° ± 3°F. and RH shall be 50 percent ± 5 percent when method B (wedge) is used, the average limit for first failure shall be not greater than 0.35 inch either along the length or width of the film.

3.8 Dimensional stability. The values for dimensional stability of the film shall not exceed the characteristics shown in table II when tested, as specified in 4.9, 4.9.1, 4.9.1.1, and 4.9.1.2.

3.9 Roll sizes. Rolls shall be wound at even tension on hollow cores. Unless otherwise specified by the procuring activity, the rolls shall be furnished in widths and lengths in accordance with

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ANSI PH-1.30. When specified (see 6.2), in the contract or purchase order, the manufacturer's standard sizes will be acceptable. Tolerance on roll widths shall be plus 3/8 inch, minus 0 inch. Tolerance on the roll length shall be plus 18 inches, minus 0 inch.

3.10 Tightness of roll wrap. Rolls of sensitized film shall be wound in such a way that the difference in minimum and maximum radii is not greater than 2-1/2 percent maximum radius when tested, as specified in 4.10.

3.11 Cuts in rolls. Rolls of sensitized film furnished by the manufacturer in lengths of 20 yards may contain 1 cut, no part of which is less than 4 yards in length. Rolls furnished by the manufacturer in lengths of 50 yards may contain 2 cuts, no part of which is less than 4 yards in length. All cuts shall be flagged at both edges with projecting colored markers. At least 1/2 yard of sensitized film shall be added to the roll for each cut. A single shipment shall contain not more than 20 percent of such cut rolls.

3.12 Sheet sizes. Unless otherwise specified (see 6.2), sheets of sensitized film shall be furnished in sizes specified in Fed. Std. No. 131 or ANSI PH-1.16. The sheets shall be cut square on four sides, and shall be cut to the sizes specified. The dimensions shall not vary more than ± 1/16 inch for sizes 8 X 10-1/2 through 15 X 18 inches, inclusive. Sheets larger than 15 X 18 inches shall not vary more than ± 3/32 inch. The sizes of the sheets and tolerances shall apply to the sensitized film when the film is in equilibrium with a temperature of 70° ± 5°F. and 55 percent ± 5 percent relative humidity. If either dimension falls within the larger tolerance, then the larger tolerance applicable to the longer side shall be applicable to the shorter side.

3.13 Finish. The sensitized film shall have a matte finish on one or both sides (except for class 4), as specified (see 1.2). The matte surface shall accept pencil or ink to yield solid lines without skipping, feathering or smudging. The surface shall have suitable erasing qualities.

3.14 Temporary curl. The temporary curl of sensitized film shall not prevent the satisfactory front and rear stacking of finished prints in machines designed for automatic front or rear stacking (see 4.12).

3.15 Permanent curl. The permanent curl of types I and II finished prints shall not exceed 12 millimeters when tested, as specified in 4.12.

3.16 Reproducible quality. All sensitized films, specified herein, shall be capable of producing a faithful reproduction of the original. The sensitized film shall be capable of producing prints with clean, even backgrounds, free from mottle, spots or other defects which would make the prints unsuitable for the purpose intended.

3.17 Diazo film characteristics. The characteristics of the diazo film shall conform to those listed in table I. For values after fading, the samples shall be faded, as specified in 4.16.

3.17.1 Speed. The speed of the diazo film shall not be less than that specified in table I when tested in accordance with 4.13.

3.18 Developing speed. Developing speed of types I and II film shall be such that a minimum of 80 percent of full development is attained in one pass through the developing section, as specified in 4.17. Type II film requiring moist processing should be developed in a solution recommended by the manufacturer.

3.19 Expiration date. Unless otherwise specified, the expiration date of the sensitized film shall be not less than 90 days from date of shipment.

3.20 Uniformity. Uniformity of physical and sensitometric characteristics among specimens of the same emulsion or coating number, and specimens of a different emulsion or coating number shall be maintained within the limits which will ensure that no significant difference of any characteristic will be noticeable between different specimens of the same product. Variations of physical and sensitometric characteristics between different batches of an emulsion or coating number, or within the same batch shall not exceed the limits set forth in this specification.

3.21 Workmanship. Workmanship shall be in accordance with the highest grade commercial practice covering the manufacture of diazotype sensitized [illegible].

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TABLE I. Requirements for types I and II

Class	Type	Style	*Min. print speed (1)	Min. actinic opacity (percent) (D-Max. Area)		Min. actinic transmission (percent) (D-Min. Area)		D-Max. (Minimum)		D-Min. (Maximum)	
				Before fading	After fading	Before fading	After fading	Before fading	After fading	Before fading	After fading
1	I	1A	20	80	70	30	25	1.25	1.10	0.30	0.40
2	I	1A	20	80	70	25	20	1.25	1.10	0.30	0.40
3	I	1A	20	80	70	30	25	1.25	1.10	0.30	0.40
4	I	1A	20	80	70	35	30	1.25	1.10	0.30	0.40
1	I	1B	20	85	70	30	25	0.95	0.85	0.30	0.40
2	I	1B	20	85	70	25	20	0.95	0.85	0.30	0.40
3	I	1B	20	85	70	30	25	0.95	0.85	0.30	0.40
4	I	1B	20	85	70	35	30	0.95	0.85	0.30	0.40
1	II	1B	20	80	70	25	20	1.10	0.95	0.35	0.45
2	II	1B	20	80	70	25	20	1.10	0.95	0.35	0.45
3	II	1B	20	80	70	30	25	1.10	0.95	0.35	0.45
4	II	1B	20	80	70	35	30	1.10	0.95	0.35	0.45

*(1) Measured in accordance with UU-P-221, (see 4.13).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.2), in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling and inspection for acceptance. Unless otherwise specified (see 6.2), sampling for inspection and acceptance shall be performed in accordance with the provisions of MIL-STD-105. For purposes of sampling, an inspection lot shall consist of all material of the same type, class and style submitted for delivery at one time. Alternate manufacturer's sampling and inspection procedures may be utilized providing such sampling and inspection methods will assure quality equal to or better than that obtained from MIL-STD-105, and provided the manufacturer's methods have been approved by the procuring activity.

4.2.1 Inspection of material and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 Inspection and quality of the end item.

4.2.2.1 Quality of the end item. The quality of the end item is the manufacturer's responsibility and can be determined by means of inspection and sampling set forth in 4.2 However, the quality of the end item must be equal to or better than the quality obtained by requirements set forth in the following paragraphs:

4.2.2.1.1 Physical examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels, and acceptance levels (AQL's) set forth below. Not more than one roll of roll film or 1 package of sheet film shall be drawn from any one shipping container for each lot, for each type, class, subclass, and style of film offered for acceptance for visual and dimensional characteristics. The lot size for purposes of determining sample size in accordance with MIL-STD-105 shall be expressed in units of rolls, or packages of sheets for examinations under 4.2.2.1.2, 4.2.2.1.3, and 4.2.2.1.4 and in units of shipping containers for examination in 4.2.2.1.5. Examination included in 4.2.2.1.2, 4.2.2.1.3, and 4.2.2.1.4 may be performed at any point immediately before, during or after spooling operation.

4.2.2.1.2 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be one sheet of sensitized film or two linear feet of sensitized roll film, as applicable. Not more than one sample unit shall be drawn from any package or roll.

Examine	Defect
Appearance and workmanship	Sheets not cut square as specified. Film not completely or evenly coated. Film containes scuffs, scratches, foreign matter, or abrasions on emulsion surface. Abnormal longitudinal curl. Holes, tears, cuts, cracks, or sharp creases.

4.2.2.1.3 Examination of the end item for dimensional defects. The sample unit for this examination shall be one sheet of film or one roll of film, as applicable. Not more than one sample unit sheet shall be drawn from one package of sheets.

Examine	Defect
Dimensions:	
Sheet Film:	
8 X 10-1/2 through 15 X 18 inches.	Length or width varies by more than plus or minus 1/16 inch from specified size.
Sizes larger than 15 X 18 inches.	Length or width varies by more than plus or minus 3/32 inch from specified size.

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Examine	Defect
Roll Film Width	Varies by more than plus 3/8 inch or minus 0 inch.
Roll Film Length	Not in accordance with 3.9.

4.2.2.1.4 Examination of the end item for defects in roll construction, blocking, and count of sheets. The sample unit for this examination shall be one roll or one package of cut sheets, as applicable.

Examine	Defect
Roll Film	Rolls not evenly or neatly wound; not wound on specified core; dimensions of core not as specified. Core cracked, broken, chipped, mutilated or collapsed. Not spooled as required. Roll contains more than permitted number of cuts. Flag at cut missing or not as specified.
Unwinding of Roll	Does not unroll evenly or uniformly. Film blocks during unwinding.
Count per package of sheets	Number of sheets per package less than specified.

4.2.2.1.5 Examination of preparation for delivery. An examination will be made to determine that marking, materials, workmanship, contents and weights comply with requirements of section five of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. The lot size shall be the number of shipping containers fully prepared for delivery, submitted for acceptance at one time.

Examine	Defect
Packaging and preservation (as applicable)	Not as specified, not in accordance with contract requirements. Flat sheets or rolls (as applicable) not unit wrapped and packaged as specified.
Packing	Unit packages not uniform. Containers of improper construction. Not packed as specified. Packaging material not as specified; closures not accomplished as specified or by required methods or materials. Number of rolls, boxes or envelopes per carton less than indicated or specified.
Markings (interior and exterior)	Data on labels, when specified, not in accordance with contract specifications. Labels not properly attached. Exterior labels or markings illegible, incorrect, incomplete, or not in accordance with contract requirements.
Weight (interior and exterior)	Gross or net weight exceeds contract requirements.

4.2.2.1.6 Inspection levels and acceptable quality levels AQL's for examination. The inspection levels, for determining the sample size and acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination	Inspection Level	AQL
4.2.2.1.1	I	1.0
4.2.2.1.2	S-3	2.5
4.2.2.1.3	S-2	2.5
4.2.2.1.4	S-2	4.0

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4.3 Testing of the end item. Testing of the end item shall be conducted in accordance with table III for the characteristics as indicated therein for each lot submitted for acceptance. The sample unit, as referenced herein, is defined as the amount of material required to perform all tests one time. The sample unit for all testing shall consist of one package of cut sheet film or one roll, as applicable. Three sample units each drawn from a different shipping container, shall be tested with no evidence of failure to meet sample unit or average requirements, as applicable. As an alternate, samples for testing for sensitometric characteristics, tensile strength, brittleness, tear resistance, and dimensional stability characteristics may be drawn at any point after the emulsion coating operation. Samples for the remaining tests shall be drawn at any point not sooner than immediately prior to the beginning of the packaging operation.

4.4 Thickness. Remove the coatings from both sides of a film sample with a suitable solvent. Using a micrometer caliper or other suitable device which is capable of measuring thickness accurately to 0.0001 inch, measure the thickness of the base film.

4.5 Coating adhesion. Not less than 5 samples of the sensitized material under test, 8-1/2 X 11 inches each, shall be fully processed in accordance with the manufacturer's recommendation. For Type I material, treat the processed sheet at 80°C. (176°F.) for five minutes to remove excess ammonia and allow to return to room temperature. The coatings of the processed samples shall not be removed when a 1/2-inch wide strip of transparent pressure sensitive tape, conforming with PPP-T-60, is placed firmly on the surface and pulled away with a sharp tug in a direction 90° from the flat plane of the sample under test.

4.6 Coating defects. One sample of each type of film specified herein, not less than 8-1/2 X 11-inch-cut sheet, nor less than one linear yard in length and the full width of roll material, shall be selected at random from the sensitized plastic film to be tested. The samples of each type shall be cut into thirds. Two thirds of all samples shall be put aside protected from exposure to light. The unprotected third shall be examined in visible light to determine conformance with the requirements specified in 3.5. One third of the protected samples shall be exposed and fully processed on applicable equipment for each type of plastic sheet. These samples (after exposure and processing) shall be held no more than 6 feet from an unshielded 60-watt incandescent lamp and again visually examined for compliance with requirements specified in 3.5. The final protected third of all samples shall be exposed using a test film consisting of a 1/4-inch grid pattern having line widths of approximately 1/16-inch and processed on applicable equipment. These samples (after exposure and processing) shall be visually examined in white light for coating streaks or other defects which would impair print quality.

4.7 Tensile strength. The test procedure used for measuring the tensile strength of the film shall be in accordance with method 1013 of Fed. Test Method Std. No. 406. Ten samples shall be cut from the lengthwise direction of the film and shall be subjected to the above test.

4.8 Brittleness. The film shall be tested for brittleness in accordance with ANSI PH-1.31.

4.9 Thermal and humidity coefficients of expansion. The humidity coefficient and the thermal coefficient shall be determined on processed film in accordance with ANSI PH-1.32, sections 6 and 7, respectively. Measurements should be made preferably by method A (pin gage procedure); although method B or C may be used. The thermal coefficient of expansion shall be determined in the range of 70°F. to 120°F. at a relative humidity of 20 percent, and the humidity coefficient of expansion shall be determined in the range of 15 to 60 percent relative humidity at 70°F.

4.9.1 Permanent change, temperature cycle. Precondition processed samples at 70° ± 3°F. and 50 ± 3 percent relative humidity for 16 hours and measure (preferably method A, ANSI PH-1.32). Place the samples in a circulating air oven maintained at 180° ± 3°F. for one hour, recondition for 16 hours at the original conditions, and remeasure. The percent change is calculated as follows:

$$\text{Percent change} = \frac{D_o - D_f \times 100}{D_o}$$

D_o = original length (or width) of test sample in inches.

D_f = final length (or width) of test sample in inches.

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TABLE II. Dimensional stability

Thermal coefficient expansion (1)	Humidity coefficient expansion (2)	Permanent change temperature	Permanent change humidity	Permanent change processing
Subclass a. 0.000015 inch	0.000020 inch	± 0.10%	± 0.05%	± 0.03%
Subclass b. 0.000020 inch	0.000025 inch	± 0.15%	± 0.010%	± 0.06%

(1) Inches per inch per degree F. - Maximum.

(2) Inches per inch per percent relative humidity - Maximum.

4.9.1.1 Permanent change, humidity cycle. Precondition processed samples at 70° ± 3°F. and 50 ± 3 percent relative humidity for 16 hours and measure (perferably method A, ANSI PH-1.32). Expose the samples to an atmosphere of 70° ± 3°F. and 90 ± 3 percent relative humidity for 24 hours then recondition under the original conditions for 16 hours and remeasure. The percent change is calculated as follows:

$$\text{Percent change} = \frac{D_o - D_f}{D_o} \times 100$$

D_o = original length (or width) of test sample in inches.

D_f = final length (or width) of test sample in inches.

4.9.1.2 Permanent change, processing cycle. The permanent change due to processing shall be determined using Procedures II and III and ANSI PH-1.32 (measurements by method A). Exposure and processing carried out in accordance with manufacturer's recommendations or, as specified in 4.13, the material under test. The percent change is calculated as follows:

$$\text{Percent change} = \frac{D_o - D_f}{D_o} \times 100$$

D_o = original length (or width) of test sample in inches.

D_f = final length (or width) of test sample in inches.

4.10 Roll tightness. The tightness of a roll of sensitized film shall be determined as follows: Suspend the roll of film on a shaft with its axis in a horizontal position. Allow the roll to remain without rotation for 1 hour in this position. The perpendicular distance from the top outer edge of the core to the upper edge of the roll and the bottom outer edge of the core to the bottom edge of the roll shall be measured to the nearest 1/32 of an inch. The difference between these two dimensions shall not be greater than 2-1/2 percent of the roll diameter.

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TABLE III. Instruction for testing (Sample Unit)

Characteristics	Specification reference		Requirements applicable to		Number determinations per sample unit	Results reported as	
	Requirement	Test method	Ind. units	Lot average		Pass or Fail	Numerically to nearest
Thickness*	3.3	4.4	X	-	Avg. of 2	-	0.0001
Coating adhesion	3.4	4.5	X	-	Avg. of 2	X	-
Coating defect	3.5	4.6	X	-	Avg. of 1	X	-
Tensile strength*	3.6	4.7	X	-	Avg. of 2	X	1 pound
Brittleness*	3.7	4.8	X	-	Avg. of 2	X	-
Fading	TABLE I	4.16	X	-	Avg. of 2	X	-
Dimensional stability*							
Permanent change Temperature cycle	TABLE II	4.9.1	-	X	Avg. of 2	X	0.01%
Humidity cycle	TABLE II	4.9.1.1	-	X	Avg. of 2	X	0.01%
Processing	TABLE II	4.9.1.2	-	X	Avg. of 2	X	0.01%
Coefficient of Expansion							
Thermal	TABLE II	4.9	-	X	Avg. of 2	X	0.000001 in.
Humidity	TABLE II	4.9	-	X	Avg. of 2	X	0.000001 in.
Roll tightness	3.10	4.10	X	-	Avg. of 2	X	-
Pencilling, inking and erasing		4.11	X	-	Avg. of 2	X	-
Temporary curl	3.14	4.12	X	-	Avg. of 2	X	-
Permanent curl	3.15	4.12	-	-	Avg. of 2	X	millimeter
Print speed	TABLE I	4.13	X	-	Avg. of 2	-	1.0 speed

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TABLE III. Instruction for testing (Sample Unit) (Continued)

Characteristics	Specification reference		Requirements applicable to		Number determinations per sample unit	Results reported as	
						Pass or Fail	Numerically to nearest
	Requirement	Test method	Ind. units	Lot average			
Actinic opacity and actinic transmission (faded and unfaded) as applicable	TABLE I	4.14, 4.15	X	-	Avg. of 2	-	% opacity or % trans.
Minimum and maximum densities (faded and unfaded) as applicable	TABLE I	4.18, 4.19	X	-	Avg. of 2	-	0.01
Developing speed (as applicable)	3.18	4.17	X	-	Avg. of 2	-	% development

*These tests shall be performed on first production lot and may be certified for succeeding lots.

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4.11 Penciling, inking and erasing tests. (As appropriate)

4.11.1 Inking test. Ten lines shall be drawn on the matte surface of an exposed, processed, and dried sample of plastic film. The ink shall conform to TT-I-528. The ink lines shall be drawn with a technical fountain pen with appropriate drawing point and shall not exceed 1/64 inch in width and not less than 5 inches in length. The ink shall adhere uniformly and the lines shall not show skipping, feathering, or spreading after drying the test sheet for 15 minutes at 100°F. A strip of transparent adhesive tape, 1/2-inch wide, and conforming to PPP-T-60 shall be placed at right angles across the ink lines and then pulled away with a sharp tug. The tape shall separate from the surface without removing the drawn ink lines, and there shall be no noticeable loss in density of the ink lines at the spot where the tape has been applied.

4.11.1.1 Ink erasing test. Prepare another sample, as specified in 4.11.1, but omit the step with the pressure sensitive tape. After 15 minutes, erase the ink lines. Ink lines shall be drawn upon the erased area at right angles to the lines originally drawn. After 15 minutes these lines shall be erased. All erasures shall be made with a moistened grit-free pencil eraser. The erased areas shall show no smudging, feathering, or ghost lines after erasure. The erasure shall not cause the matte surface to burnish or gloss such as to provide reflective qualities detrimental to reproductions.

4.11.1.2 Penciling test. Ten lines shall be drawn on the matte side of an exposed, processed, and dried sample. The lines shall be not less than 5 inches in length and shall be drawn with a 3H pencil lead and a 3H plastic pencil lead sharpened to 0.10-inch diameter conical point. The sample will then be examined to determine that the pencil lines are solid and opaque.

4.11.1.3 Pencil erasing test. Pencil lines shall be capable of being erased with a grit-free eraser without leaving ghost or trace of lines after erasing. There shall be no noticeable loss of pencil take after three erasures performed on the same areas nor any loss in opacity of solid pencil lines. Erasures shall not cause the matte finish to burnish or gloss such as to provide reflective qualities detrimental to reproductions.

4.12 Curl. On a commercial diazotype printer-processor, in good working condition, adjust the speed and developing conditions to the setting normally used for the product under test when printing from an intermediate master. Let the machine run for one hour to allow the developing conditions to come to equilibrium. (On starting up a moist diazotype printer-processor, let it warm up for an additional one half hour.) Expose and develop 25 sheets, each 8-1/2 X 11 inches, of the material under test in immediate succession using an intermediate master. In so doing, direct the sheets into the machine so the longest dimension is parallel to the direction of travel. To be satisfactory, all prints must stack without requiring manual attention. In a conditioning chamber, capable of maintaining a relative humidity of 50 percent ± 2 percent and a temperature of 73° ± 3.5°F., place two sheets, selected at random from the ones tested above, on a flat surface, one print side up and the other print side down. Allow the prints to remain in the chamber at approximately 73°F. and 50 percent relative humidity for a period of 72 hours. At the end of this period, measure and record in millimeters the greatest lift from the flat surface of any part of either sheet. Record the effects of temporary curl in terms of stacking performances; "Stacks Satisfactorily" or "Stacks Unsatisfactorily". Report the permanent curl as millimeter curl.

4.13 Print speed, types I and II. The print speed shall be determined, as specified in UU-P-221. All exposures and development for types I and II material shall be made on the apparatus specified in UU-P-221, as applicable.

4.14 Actinic opacity, D-Max area. The actinic opacity of the maximum density area shall be measured in accordance with the method specified in UU-P-221.

4.15 Actinic transmission, D-Min area. The actinic transmission of the minimum density area shall be measured in accordance with the method specified in UU-P-221.

4.16 Fading test. The fading of the test prints shall be in accordance with the method specified in UU-P-221.

4.17 Development at print speed. Completely develop, without exposure, a sheet of material under test, not less than 8-1/2 X 11 inches, on a commercial diazotype printer-processor in good working condition. The speed of the printer-processor should be such that it is correct for exposing the material using a master with an actinic transmission of 45 to 55 percent. Adjust the processor to produce optimum developing conditions and wait one hour for the developing section to come to equilibrium. (For type II wait an additional one-half hour.) Send the test sheet through the developing section of the diazotype printer-processor one time at the print speed as determined above. Cut the developed sheet in half immediately after development and send one-half through the exposing

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section of the printer-processor at one-half the print speed to burn out all undeveloped diazo and prevent further development. Send the other half of the sheet again through the developing section at one-half the print speed to assure full development. Take five reflection density readings across the diagonal on the D-Max side of the sheet that was developed once. For this test use an opaque white backing sheet with a reflection density of 0.07 ± 0.005. Determine and record the average of the readings as the full density. Calculate the development at print speed by:

$$\frac{\text{One-pass density} \times 100\%}{\text{Full density}} = \text{development at print speed full density}$$

A densitometer, with zero and sensitivity settings checked with appropriate working standards, shall be used. The density shall conform to ANSI PH-2.17. The densitometer shall be equipped with S-4 response phototube, a Wratten 106 filter and an IR 4 filter. The densitometer shall be adjusted to read "0" for 100 percent reflectance. This may be accomplished by using a fresh magnesium oxide surface, (see NBS LC-547) which if properly prepared, will have a reflectance of 100 percent for such a densitometer. A more convenient standard, which has been calibrated against magnesium oxide, with a reflectance between 78 to 100 percent (density 0.00 to 0.108) may be used for setting the low end of the scale. A calibrated standard for approximately one percent reflectance (density 2.00) shall be used for checking the high density end of the scale.

4.18 Maximum density. Completely develop without exposure a sheet of material under test of at least 8-1/2 X 11 inches. For type I material, heat the developed sheet for five minutes at 80°C. (176°F.) to remove excess ammonia and allow to cool to room temperature before proceeding. Check the zero and sensitivity settings of the densitometer (as described in 4.17) with the appropriate working standards. Take five reflection density readings across the diagonal on the image side of the sheet. For this test use an opaque white backing sheet with a reflection density of 0.07 ± 0.005. Determine and record the average of the readings as the maximum density.

4.19 Minimum density. Fully expose and develop at a speed that will assure complete burnout, a sample of at least 8-1/2 X 11 inches. Check the zero and sensitivity settings of the densitometer (same as described in 4.17). Take five reflection density readings across the diagonal on the image side of the sheet. For this test use an opaque white backing sheet with a reflection density of 0.07 ± 0.005. Determine and record the average of the readings as the minimum density.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Sheet or roll film, in quantities, as specified (see 6.2), shall be wrapped in water-vaporproof barrier material, and sealed. Stiffener boards shall be placed on top and bottom of the sheet film to prevent bending. When specified (see 6.2), each wrapped package shall be packaged in a close-fitting box conforming to PPP-B-636, class domestic, style optional. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.1.2 Level C. Sheet or roll film, in quantities, as specified (see 6.2), shall be packaged in accordance with the manufacturer's standard practice, providing that this insures protection for the film during shipment and safe delivery to its destination.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Film of like description packaged, as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in close-fitting boxes conforming to PPP-B-621, class 2; PPP-B-636, class weather-resistant or to PPP-B-640, class 2. When the PPP-B-621 boxes are utilized, each box shall be provided with a waterproof case liner conforming to MIL-L-10547. The PPP-B-621 boxes shall be strapped in accordance with the appendix to PPP-B-621, the PPP-B-636 and PPP-B-640 boxes shall be closed, waterproof sealed and strapped in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Film of like description packaged, as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in close-fitting boxes conforming to PPP-B-636, class domestic or to PPP-B-640, class 1. The boxes shall be closed in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Film of like description packaged, as specified in 5.1, shall be packed in quantities, as specified (see 6.2), in containers that will assure safe arrival at destination in compliance with the National Motor Freight Classification Rules or the Uniform Freight Classification Rules, as applicable.

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5.3 Palletization. When specified (see 6.2), film packed, as specified in 5.2, shall be palletized in accordance with MIL-STD-147.

5.4 Marking. In addition to the special marking required by 5.4.3, and any markings required by the contract or order, all marking shall be in accordance with 5.4.1 or 5.4.2, as specified (see 6.2).

5.4.1 Civil agencies. All marking shall be in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. All marking shall be in accordance with MIL-STD-129.

5.4.3 Special marking. In addition to the markings required by 5.4.1 or 5.4.2. the following precautionary marking shall be placed on all interior packages and shipping containers:

"FILM, DIAZOTYPE
STORE IN A COOL, DRY LOCATION.
PROTECT FROM AMMONIA. AVOID EXCESSIVE EXPOSURE TO WHITE
FLORESCENT LIGHT OR DAYLIGHT."

6. NOTES

6.1 Intended use. The sentized film covered by this specification is intended for use in making intermediates and reproductions of original copy, drawings and tracings.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.

(b) Type, class, subclass, and style desired (see 1.2).

(c) Roll or sheet, state which (see 3.1).

(d) Date when required (see 3.2 and table III).

(e) Thickness (see 3.3).

(f) Roll width and length (see 3.9).

(g) Size and number of sheets per box (see 3.12).

(h) Responsibility for inspection, if other than specified (see 4.1).

(i) Selection of samples, if other than specified (see 4.2).

(j) Selection of level of packaging and packing required (see 5.1 and 5.2).

(k) Marking, if other than specified (see 5.4).

(l) Palletization, unitization or containerization (see 5.3).

6.3 Cross reference of classification. The title of this specification has been changed by deletion of the words, "Wash off Process, Brownprint." The types, subtypes and classes have been changed as indicated below. The items covered by this specification may be binned with prior diazo items.

Basic Specification	This Specification
Subtypes:	
A - Wash-off (tannable colloid)	Deleted
B - Diazotype	Deleted
C - Brownprint (Iron silver)	Deleted

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Basic Specification	This Specification
Styles:	
1A - Black line	1A - Blackline (Diazotype - Print film)
1B - Sepia line	1B - Diazotype - Intermediate film

6.4 For civil agency procurement. When level B packaging is required for civil agency procurement, the level A requirements in 5.1.2.1 or 5.1.2.2 shall apply.

MILITARY CUSTODIANS:

Army - EL
Navy - AS
Air Force - 70

Review activities:

Army - EL
Navy - AS
Air Force - 70, 85

User activities:

Army - MB, AT

Preparing activity:

Navy - AS

Civil Agency Coordination Activity:

GSA-FSS

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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